

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Semester Examination – 2024

Course: Bachelor of Pharmacy

Semester: VII

Subject Name: Instrumental Method of Analysis

Subject Code: BP701T

Max Marks: 75

Date: 18/12/2024

Duration: 3 Hrs.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Que 1) Objective Type Questions (Answer All the Questions)

(2 X 10) = 20

- i) Define chromatography & classify chromatographic technique.
- ii) Explain Chromophore & Auxochrome.
- iii) Enlist the detector used in gas chromatography.
- iv) How paper chromatography differentiated from TLC?
- v) Define quenching & enlist different types of quenching.
- vi) Write the principle & application of UV-Visible spectroscopy.
- vii) Write short notes on electronic transition.
- viii) Write the application of electrophoresis.
- ix) Write notes on photomultiplier tube.
- x) Write difference between nephelometry & turbidometry.

Que 2) Long Answer Question (Answer 2 out of 3)

(10 X 2) = 20

- i) Write principle, theory, stationary phase used & application of HPLC techniques.
- xi) Write instrumentation & application of Ion Exchange Chromatography. Write notes on exchanger.
- ii) Write instrumentation & application of IR spectrometer. Explain the detector used in IR spectrometer.

Que 3) Short Answer Question (Answer 7 out of 9)

(5 X 7) = 35

- i) Explain in detail about Phototubes & photoemission cells.
- ii) Discuss about principle & instrument of column chromatography.
- iii) Write principle & steps involved in TLC technique.
- iv) Explain Beer's & Lambert's law with their deviation.
- v) Write notes on factor affecting vibration in IR.
- vi) Write about methodology & application of paper chromatography.
- vii) Write notes on detector used in flame photometry.
- viii) Write instrumentation & application of Ion Exchange Chromatography.
- ix) Write details about Quenching. Explain the factor affecting fluorescence.

*** END OF THE PAPER***